

Ganesh Gupta

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EXPERIENCE

Oracle | Software Engineer, AI Platform

Jun 2025 – Present

- Implemented **GraphRAG** (knowledge-graph-augmented retrieval) in production, improving grounding accuracy over standard vector retrieval for complex, **multi-hop** queries.
- Designed deployment-gated **evaluation frameworks** translating abstract quality objectives into measurable, structured metrics, running **A/B tests** to validate model changes before full rollout and blocking 15% of failed releases before production.

QureAI | ML Infrastructure Engineer

Mar 2026 – Present

- Designed and implemented a **Neo4j knowledge graph** modeling domain entities and relationships, powering **graph-based** retrieval and reasoning in a production pipeline.
- Rebuilt the **ranking** pipeline with **hybrid dense-sparse retrieval** and **re-ranking**, improving recall **61% to 89%** on a 2K-query eval set while sustaining **p99 latency of 140ms** across **50K+ daily requests**.

University of Texas at Arlington | Graduate Teaching Assistant

Aug 2024 – May 2025

- Benchmarked 5+ **multi-agent reinforcement learning** architectures on a multi-GPU **PyTorch** cluster, evaluating **reward design** and robustness under distributed interference.
- Developed multi-modal deep learning models in **PyTorch** with **attention mechanisms** and domain-specific architectures, building **evaluation frameworks** and reproducible research workflows.

Nomura Research Institute Financial Technologies | Software Engineer

Sep 2019 – Aug 2023

- Architected and operated **25+ production microservices** for **1M+ users**, sustaining **99.9% uptime** over 4 years, standardizing failure-handling contracts that cut production escalations **60%** in one quarter.

PROJECTS

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent with **Pinecone** vector retrieval and **hybrid re-ranking**, decomposing natural-language queries into dependency-ordered subqueries with structured relevance scoring.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Eval platform with **rubric-based scoring**, hallucination detection, and drift tracking, translating subjective quality judgments into structured, measurable metrics across model versions.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent** system automating production incident diagnosis via **tool-use** chains, routing inputs to specialized sub-agents and synthesizing root cause reports.

EDUCATION

University of Texas at Arlington | M.S. Computer Science, GPA 4.0/4.0

Aug 2023 – May 2025

MCKV Institute of Engineering | B.Tech Computer Science

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Java, C++, TypeScript, SQL

ML & Research: PyTorch, Reinforcement Learning, Attention Mechanisms, Representation Learning, A/B Testing

Ranking & Retrieval: Hybrid Dense-Sparse Retrieval, Re-Ranking, Vector Search (Pinecone), Knowledge Graphs (Neo4j), GraphRAG

Evaluation & Systems: Eval Frameworks, Observability, Real-Time Low-Latency Serving, AWS, GCP, Docker, Kubernetes

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer